

Laboratory Session

Signal Conditioning, Characterization, and Visualization

Learning Objectives

By the end of this session, students will:

- Identify the function and application of differential and non-inverting amplifier circuits in sensor signal conditioning.
- Understand how to characterize a sensor through calibration and response analysis.
- Design and implement a voltage comparator to trigger a visual indicator.
- Create a basic virtual instrument (VI) for data acquisition and visualization.
- Analyze the performance and limitations of a complete measurement chain for force sensing using a load cell.

Materials

Device/component	Quantity
Multimeter	1
Power source	1
Load cell	1
Oscilloscope (2 probes)	1
UA741 OpAmp	4
ESP32	1
Breadboard, wire, resistors and capacitors	
Different known weights in the load cell range	

Pre-Lab Setup

All circuits must be pre-assembled before the start of the session both in Multisim Live and breadboard:

- Differential Amplifier (gain ≈ 10)
- Non-Inverting Amplifier (adjustable gain)
- Passive Low-Pass Filter (RC)
- Voltage Comparator

Read the following:

- <https://predictabledesigns.com/introduction-to-load-cell-conditioning-circuits/>
- https://www.electronics-tutorials.ws/opamp/opamp_5.html
- <https://www.electronics-tutorials.ws/opamp/op-amp-comparator.html>

Watch and implement the VI from the following video:

- <https://www.youtube.com/watch?v=MrqQF7eGJK0>

Procedure

Load cell conditioning circuit

1. Connect the load cell's output to a differential amplifier with a gain of 10 to amplify the millivolt-level signal.
2. Use the output of the differential stage as input to a non-inverting amplifier. Adjust the gain so that the final output voltage remains under 3 V at full load.
3. Visualize the output on the oscilloscope and determine a suitable RC passive low-pass filter to smooth the signal (use FFT tool).
4. Implement a voltage comparator that turns on an LED when the force exceeds the maximum calibrated threshold.

Data acquisition

1. Characterize the ADC from the ESP32 using at least 10 different known voltage values.
2. Use the ESP32 to read the filtered analog signal (via ADC).
3. Transmit the ADC values via UART to a customized LabVIEW VI for visualization.
4. Characterize the system using known weights, record the output values, and create a calibration curve (force vs. voltage). Consider using cftool.
5. Given the characterization, program the microcontroller so that it also sends the force value.

Report requirements

Your report should cover the following:

- A clear schematic of the full measurement chain (from load cell to visualization).
- Plot of the ADC-voltage curve.
- Calibration data: force vs. voltage with regression analysis.
- Graphs showing system responses before and after filtering.
- Screenshots of oscilloscope signals and simulation outputs.
- Brief documentation of your LabVIEW VI (interface, data acquisition flow).
- Highlight how the signal conditioning stages affect resolution and accuracy.

Analytical Reflections (don't write them as Q&A in your report, you must develop them in your discussion)

- How effective was the gain staging in adapting the load cell signal to the ADC input range?
- How does the low-pass filter affect the sensor's response to rapid force changes?
- How does the comparator improve system usability or safety?
- What are the static characteristics (e.g., sensitivity, linearity) of your setup?
- How does each conditioning stage influence the fidelity of the measured signal?
- Compare simulation vs. experimental results for each stage.
- What design choices would you make differently for a higher-speed or higher-resolution application?
- How does the system behave under noisy or unstable load conditions?